

Supplementary Information

An Accuracy and Calibration Benchmark of scGPT Pipelines and Matched Classical Models for Cell Type Annotation

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Contents and scope

This document reports the complete numerical outputs underlying the revised manuscript. It includes all 36 within-atlas pipelines, paired differences, every donor-fold result, donor- and cell-bootstrap comparisons, all label-scarcity settings, temperature-scaling results, computational identifiers, and reliability diagrams with bin counts and uncertainty. Results derived from the superseded custom wrapper, end-to-end fine-tuning, or embedding-based gene-similarity analysis are not included.

- Supplementary Table S1: all within-atlas discrimination and calibration metrics.
- Supplementary Table S2: paired within-atlas Macro F1 differences.
- Supplementary Table S3: all five donor-held-out folds and six pipelines.
- Supplementary Table S4: donor versus cell uncertainty and paired donor differences.
- Supplementary Table S5: complete label-scarcity results.
- Supplementary Table S6: complete temperature-scaling results.
- Supplementary Tables S7–S8: dataset, checkpoint, and software audit.
- Supplementary Figures S1–S6: within-atlas reliability diagrams.
- Supplementary Figure S7: donor-held-out reliability diagrams.

Supplementary Table S1: Complete within-atlas results for all 36 representation–classifier pipelines.

Dataset	Pipeline	Macro F1	95% CI	Accuracy	Bal. acc.	ECE	Adapt.	Classwise	Brier	NLL
TNBC	scGPT emb. + LR	0.978	0.977–0.980	0.974	0.974	0.010	0.010	0.003	0.040	0.081
TNBC	scGPT emb. + RF	0.968	0.966–0.969	0.965	0.952	0.043	0.043	0.014	0.061	0.138
TNBC	scGPT emb. + XGB	0.980	0.979–0.981	0.976	0.975	0.001	0.002	0.001	0.036	0.069
TNBC	Matched expr. + LR	0.991	0.990–0.992	0.989	0.990	0.002	0.002	0.001	0.017	0.034
TNBC	Matched expr. + RF	0.987	0.986–0.989	0.986	0.981	0.032	0.032	0.010	0.027	0.067
TNBC	Matched expr. + XGB	0.992	0.991–0.992	0.990	0.990	0.002	0.002	0.001	0.015	0.030
Indonesia PBMC	scGPT emb. + LR	0.929	0.926–0.932	0.924	0.916	0.021	0.021	0.004	0.114	0.210
Indonesia PBMC	scGPT emb. + RF	0.912	0.909–0.915	0.906	0.893	0.074	0.074	0.012	0.154	0.293
Indonesia PBMC	scGPT emb. + XGB	0.936	0.933–0.938	0.924	0.930	0.004	0.004	0.001	0.112	0.200
Indonesia PBMC	Matched expr. + LR	0.956	0.953–0.958	0.953	0.957	0.009	0.008	0.001	0.072	0.133
Indonesia PBMC	Matched expr. + RF	0.929	0.925–0.933	0.944	0.903	0.104	0.104	0.016	0.109	0.228
Indonesia PBMC	Matched expr. + XGB	0.955	0.953–0.958	0.952	0.953	0.004	0.005	0.001	0.070	0.124
Brain atlas	scGPT emb. + LR	0.871	0.831–0.902	0.984	0.812	0.025	0.025	0.006	0.029	0.082
Brain atlas	scGPT emb. + RF	0.893	0.857–0.920	0.986	0.835	0.026	0.027	0.006	0.029	0.092
Brain atlas	scGPT emb. + XGB	0.931	0.904–0.953	0.988	0.895	0.008	0.008	0.002	0.021	0.060
Brain atlas	Matched expr. + LR	0.945	0.925–0.962	0.985	0.937	0.008	0.008	0.002	0.025	0.069
Brain atlas	Matched expr. + RF	0.831	0.790–0.865	0.984	0.775	0.030	0.030	0.007	0.029	0.084
Brain atlas	Matched expr. + XGB	0.952	0.929–0.970	0.988	0.924	0.005	0.005	0.001	0.019	0.043
Multi-tissue TME	scGPT emb. + LR	0.321	0.318–0.325	0.493	0.306	0.016	0.017	0.006	0.667	1.500
Multi-tissue TME	scGPT emb. + RF	0.334	0.325–0.345	0.500	0.306	0.056	0.055	0.012	0.660	1.568
Multi-tissue TME	scGPT emb. + XGB	0.365	0.359–0.372	0.506	0.336	0.010	0.010	0.004	0.644	1.442
Multi-tissue TME	Matched expr. + LR	0.357	0.351–0.364	0.501	0.330	0.012	0.013	0.007	0.655	1.463
Multi-tissue TME	Matched expr. + RF	0.391	0.378–0.404	0.534	0.359	0.024	0.024	0.007	0.615	1.514
Multi-tissue TME	Matched expr. + XGB	0.416	0.404–0.430	0.532	0.379	0.023	0.023	0.007	0.614	1.368
Human pancreas	scGPT emb. + LR	0.722	0.717–0.727	0.977	0.713	0.050	0.050	0.028	0.044	0.110
Human pancreas	scGPT emb. + RF	0.743	0.720–0.784	0.979	0.730	0.045	0.045	0.024	0.043	0.099
Human pancreas	scGPT emb. + XGB	0.874	0.817–0.920	0.989	0.843	0.005	0.005	0.003	0.016	0.035
Human pancreas	Matched expr. + LR	0.983	0.962–0.999	0.998	0.987	0.001	0.001	0.001	0.003	0.005
Human pancreas	Matched expr. + RF	0.951	0.912–0.979	0.995	0.919	0.038	0.038	0.020	0.015	0.052
Human pancreas	Matched expr. + XGB	0.982	0.962–0.997	0.997	0.996	0.001	0.001	0.001	0.005	0.009
Pig pancreas	scGPT emb. + LR	0.601	0.583–0.619	0.941	0.545	0.044	0.044	0.037	0.088	0.187
Pig pancreas	scGPT emb. + RF	0.596	0.577–0.614	0.938	0.536	0.052	0.051	0.038	0.097	0.198
Pig pancreas	scGPT emb. + XGB	0.865	0.810–0.907	0.972	0.796	0.011	0.011	0.010	0.043	0.087
Pig pancreas	Matched expr. + LR	0.993	0.989–0.997	0.998	0.996	0.002	0.001	0.001	0.004	0.011
Pig pancreas	Matched expr. + RF	0.947	0.904–0.977	0.996	0.912	0.022	0.022	0.012	0.011	0.031
Pig pancreas	Matched expr. + XGB	0.991	0.987–0.995	0.996	0.992	0.002	0.002	0.002	0.006	0.011

ECE, equal-width ten-bin top-label expected calibration error; Adapt., adaptive ten-bin ECE; Classwise, classwise ten-bin ECE; Brier, multiclass Brier score; NLL, negative log-likelihood. Intervals are 5,000-draw cell-bootstrap percentile intervals conditional on the fitted model and fixed training partition.

Supplementary Table S2: Paired within-atlas Macro F1 differences for matched expression minus frozen scGPT embeddings.

Dataset	Comparison	Pipelines	Difference	95% CI	$P_{boot}(> 0)$	Draws
TNBC	Logistic Regression	Matched expr. + LR vs scGPT emb. + LR	0.013	0.011–0.014	1.000	5,000
TNBC	Random Forest	Matched expr. + RF vs scGPT emb. + RF	0.020	0.018–0.022	1.000	5,000
TNBC	XGBoost	Matched expr. + XGB vs scGPT emb. + XGB	0.012	0.010–0.013	1.000	5,000
Indonesia PBMC	Logistic Regression	Matched expr. + LR vs scGPT emb. + LR	0.027	0.024–0.030	1.000	5,000
Indonesia PBMC	Random Forest	Matched expr. + RF vs scGPT emb. + RF	0.017	0.013–0.020	1.000	5,000
Indonesia PBMC	XGBoost	Matched expr. + XGB vs scGPT emb. + XGB	0.019	0.017–0.022	1.000	5,000
Indonesia PBMC	Best observed pipeline	Matched expr. + LR vs scGPT emb. + XGB	0.020	0.017–0.023	1.000	5,000
Brain atlas	Logistic Regression	Matched expr. + LR vs scGPT emb. + LR	0.075	0.044–0.111	1.000	5,000
Brain atlas	Random Forest	Matched expr. + RF vs scGPT emb. + RF	-0.062	-0.096–0.028	0.000	5,000
Brain atlas	XGBoost	Matched expr. + XGB vs scGPT emb. + XGB	0.021	-0.001–0.044	0.968	5,000
Multi-tissue TME	Logistic Regression	Matched expr. + LR vs scGPT emb. + LR	0.036	0.030–0.042	1.000	5,000
Multi-tissue TME	Random Forest	Matched expr. + RF vs scGPT emb. + RF	0.057	0.049–0.069	1.000	5,000
Multi-tissue TME	XGBoost	Matched expr. + XGB vs scGPT emb. + XGB	0.051	0.039–0.065	1.000	5,000
Human pancreas	Logistic Regression	Matched expr. + LR vs scGPT emb. + LR	0.261	0.239–0.277	1.000	5,000
Human pancreas	Random Forest	Matched expr. + RF vs scGPT emb. + RF	0.208	0.155–0.250	1.000	5,000
Human pancreas	XGBoost	Matched expr. + XGB vs scGPT emb. + XGB	0.109	0.061–0.164	1.000	5,000
Human pancreas	Best observed pipeline	Matched expr. + LR vs scGPT emb. + XGB	0.109	0.064–0.166	1.000	5,000
Pig pancreas	Logistic Regression	Matched expr. + LR vs scGPT emb. + LR	0.392	0.374–0.411	1.000	5,000
Pig pancreas	Random Forest	Matched expr. + RF vs scGPT emb. + RF	0.351	0.306–0.386	1.000	5,000
Pig pancreas	XGBoost	Matched expr. + XGB vs scGPT emb. + XGB	0.126	0.085–0.182	1.000	5,000
Pig pancreas	Best observed pipeline	Matched expr. + LR vs scGPT emb. + XGB	0.128	0.087–0.184	1.000	5,000

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Each draw resampled the joint true-label and paired-prediction state, preserving pairing. $P_{boot}(> 0)$ is the descriptive fraction of bootstrap differences above zero, not a frequentist P value.

Supplementary Table S3: Complete fold-level donor-held-out results on Indonesia PBMC.

Fold	Pipeline	Genes	Train cells	Test cells	Train donors	Test donors	F1	Accuracy	Bal. acc.	ECE	Brier	NLL
1	scGPT emb. + LR	1,135	368,104	93,930	159	40	0.929	0.924	0.916	0.021	0.115	0.211
1	scGPT emb. + RF	1,135	368,104	93,930	159	40	0.912	0.904	0.891	0.071	0.155	0.294
1	scGPT emb. + XGB	1,135	368,104	93,930	159	40	0.934	0.924	0.928	0.005	0.113	0.202
1	Matched expr. + LR	1,135	368,104	93,930	159	40	0.956	0.952	0.955	0.010	0.073	0.136
1	Matched expr. + RF	1,135	368,104	93,930	159	40	0.927	0.944	0.900	0.104	0.110	0.230
1	Matched expr. + XGB	1,135	368,104	93,930	159	40	0.958	0.952	0.954	0.004	0.071	0.126
2	scGPT emb. + LR	1,141	369,759	92,275	161	38	0.923	0.921	0.911	0.021	0.118	0.217
2	scGPT emb. + RF	1,141	369,759	92,275	161	38	0.910	0.903	0.891	0.072	0.158	0.301
2	scGPT emb. + XGB	1,141	369,759	92,275	161	38	0.929	0.920	0.922	0.006	0.118	0.212
2	Matched expr. + LR	1,141	369,759	92,275	161	38	0.954	0.950	0.954	0.009	0.075	0.139
2	Matched expr. + RF	1,141	369,759	92,275	161	38	0.920	0.941	0.892	0.109	0.116	0.241
2	Matched expr. + XGB	1,141	369,759	92,275	161	38	0.951	0.948	0.946	0.005	0.075	0.133
3	scGPT emb. + LR	1,138	368,718	93,316	157	42	0.926	0.921	0.910	0.023	0.118	0.217
3	scGPT emb. + RF	1,138	368,718	93,316	157	42	0.905	0.898	0.881	0.078	0.165	0.316
3	scGPT emb. + XGB	1,138	368,718	93,316	157	42	0.930	0.921	0.921	0.003	0.117	0.210
3	Matched expr. + LR	1,138	368,718	93,316	157	42	0.954	0.951	0.954	0.009	0.074	0.137
3	Matched expr. + RF	1,138	368,718	93,316	157	42	0.922	0.943	0.892	0.108	0.113	0.237
3	Matched expr. + XGB	1,138	368,718	93,316	157	42	0.955	0.951	0.949	0.003	0.072	0.128
4	scGPT emb. + LR	1,132	372,019	90,015	161	38	0.930	0.927	0.914	0.022	0.110	0.203
4	scGPT emb. + RF	1,132	372,019	90,015	161	38	0.916	0.910	0.896	0.069	0.147	0.280
4	scGPT emb. + XGB	1,132	372,019	90,015	161	38	0.940	0.929	0.933	0.004	0.106	0.190
4	Matched expr. + LR	1,132	372,019	90,015	161	38	0.960	0.955	0.957	0.007	0.068	0.128
4	Matched expr. + RF	1,132	372,019	90,015	161	38	0.924	0.946	0.896	0.105	0.107	0.225
4	Matched expr. + XGB	1,132	372,019	90,015	161	38	0.958	0.954	0.952	0.004	0.067	0.119
5	scGPT emb. + LR	1,134	369,536	92,498	158	41	0.931	0.928	0.918	0.022	0.108	0.200
5	scGPT emb. + RF	1,134	369,536	92,498	158	41	0.916	0.909	0.896	0.070	0.147	0.283
5	scGPT emb. + XGB	1,134	369,536	92,498	158	41	0.937	0.928	0.930	0.003	0.106	0.190
5	Matched expr. + LR	1,134	369,536	92,498	158	41	0.958	0.955	0.958	0.008	0.068	0.126
5	Matched expr. + RF	1,134	369,536	92,498	158	41	0.929	0.947	0.901	0.102	0.105	0.221
5	Matched expr. + XGB	1,134	369,536	92,498	158	41	0.954	0.953	0.949	0.005	0.068	0.120

Supplementary Table S4: Donor-cluster uncertainty and paired donor-bootstrap comparisons.

Pipeline	F1	Cell CI	Donor CI	Donor ECE CI	Donor NLL CI
scGPT emb. + LR	0.928	0.926–0.929	0.924–0.931	0.020–0.023	0.201–0.219
scGPT emb. + RF	0.912	0.910–0.913	0.907–0.916	0.070–0.075	0.284–0.307
scGPT emb. + XGB	0.934	0.933–0.935	0.931–0.937	0.003–0.005	0.191–0.210
Matched expr. + LR	0.956	0.955–0.958	0.954–0.958	0.008–0.010	0.126–0.140
Matched expr. + RF	0.925	0.923–0.927	0.921–0.928	0.103–0.109	0.222–0.240
Matched expr. + XGB	0.955	0.954–0.956	0.953–0.958	0.004–0.005	0.119–0.132
Paired Macro F1 differences					
Matched pipeline	scGPT pipeline	Difference	Donor-bootstrap 95% CI		$P_{boot}(> 0)$
Matched expr. + LR	scGPT emb. + LR	0.029	0.027–0.031		1.000
Matched expr. + RF	scGPT emb. + RF	0.013	0.010–0.015		1.000
Matched expr. + XGB	scGPT emb. + XGB	0.021	0.019–0.023		1.000
Matched expr. + LR	scGPT emb. + XGB	0.022	0.021–0.024		1.000

All intervals use 5,000 percentile-bootstrap draws. Donor resampling carries all cells from a selected donor together.

Supplementary Table S5: Complete donor-held-out label-scarcity results.

Labels/class	Representation	Classifier	F1 mean	F1 SD	ECE mean	ECE SD	Brier mean	Brier SD	NLL mean	NLL SD
10	Matched expression	LR	0.856	0.014	0.065	0.006	0.259	0.015	0.505	0.028
10	scGPT embeddings	LR	0.679	0.013	0.537	0.026	0.834	0.001	2.103	0.007
10	Matched expression	XGB	0.643	0.032	0.089	0.014	0.461	0.042	1.031	0.091
10	scGPT embeddings	XGB	0.617	0.032	0.070	0.012	0.483	0.037	1.102	0.093
25	Matched expression	LR	0.893	0.002	0.014	0.001	0.183	0.008	0.348	0.015
25	scGPT embeddings	LR	0.738	0.011	0.532	0.021	0.726	0.002	1.706	0.007
25	Matched expression	XGB	0.781	0.010	0.035	0.008	0.270	0.010	0.565	0.020
25	scGPT embeddings	XGB	0.729	0.015	0.048	0.003	0.342	0.012	0.735	0.026
50	Matched expression	LR	0.905	0.006	0.016	0.003	0.160	0.010	0.304	0.019
50	scGPT embeddings	LR	0.784	0.008	0.494	0.006	0.614	0.003	1.377	0.007
50	Matched expression	XGB	0.836	0.007	0.021	0.002	0.203	0.011	0.410	0.023
50	scGPT embeddings	XGB	0.779	0.011	0.055	0.004	0.288	0.009	0.598	0.022
100	Matched expression	LR	0.913	0.003	0.029	0.001	0.145	0.005	0.285	0.011
100	scGPT embeddings	LR	0.817	0.010	0.430	0.007	0.497	0.006	1.081	0.012
100	Matched expression	XGB	0.872	0.005	0.020	0.003	0.160	0.007	0.317	0.014
100	scGPT embeddings	XGB	0.816	0.012	0.054	0.004	0.245	0.014	0.502	0.028
250	Matched expression	LR	0.914	0.002	0.043	0.002	0.140	0.004	0.305	0.010
250	scGPT embeddings	LR	0.848	0.009	0.323	0.005	0.365	0.006	0.779	0.012
250	Matched expression	XGB	0.897	0.003	0.027	0.001	0.136	0.005	0.266	0.009
250	scGPT embeddings	XGB	0.851	0.010	0.055	0.003	0.212	0.009	0.428	0.021
500	Matched expression	LR	0.915	0.003	0.050	0.002	0.137	0.006	0.330	0.015
500	scGPT embeddings	LR	0.863	0.008	0.246	0.005	0.290	0.008	0.615	0.013
500	Matched expression	XGB	0.911	0.003	0.029	0.002	0.121	0.005	0.237	0.009
500	scGPT embeddings	XGB	0.869	0.008	0.053	0.004	0.190	0.010	0.383	0.022

Five seed-level fits were first averaged within each outer fold; mean and sample standard deviation were then calculated across five donor folds.

Supplementary Table S6: Complete held-out-donor temperature-scaling results.

Pipeline	State	T mean	T SD	F1 mean	F1 SD	ECE mean	ECE SD	Brier mean	Brier SD	NLL mean	NLL SD
Matched expr. + LR	Uncalibrated	1.233	0.016	0.956	0.003	0.0102	0.0008	0.0730	0.0031	0.1379	0.0057
Matched expr. + LR	Calibrated	1.233	0.016	0.956	0.003	0.0022	0.0008	0.0725	0.0030	0.1340	0.0053
scGPT emb. + LR	Uncalibrated	0.806	0.012	0.925	0.003	0.0240	0.0015	0.1165	0.0046	0.2151	0.0081
scGPT emb. + LR	Calibrated	0.806	0.012	0.925	0.003	0.0030	0.0013	0.1152	0.0048	0.2065	0.0085

One temperature was fitted per fold using an inner donor split from outer-training donors. Reported values are means and sample standard deviations across outer folds.

Supplementary Table S7: Dataset and shared-input audit.

Dataset	Source cells	Final cells	Classes	Source genes	Matched genes	Zero-input exclusions
TNBC	427,823	427,802	7	32,354	1,165	21
Indonesia PBMC	462,034	462,034	13	36,030	1,135	0
Brain atlas	75,583	75,580	10	35,467	939	3
Multi-tissue TME	391,963	390,582	11	44,649	1,067	1,381
Human pancreas	26,474	26,474	4	30,172	1,149	0
Pig pancreas	22,056	22,056	4	15,487	1,104	0

Note. Source genes are variables in the public H5AD object. Matched genes are the primary-split training-selected highly variable genes remaining after vocabulary matching. Donor folds used independently selected sets of 1,135–1,141 genes. All source labels had more than five cells.

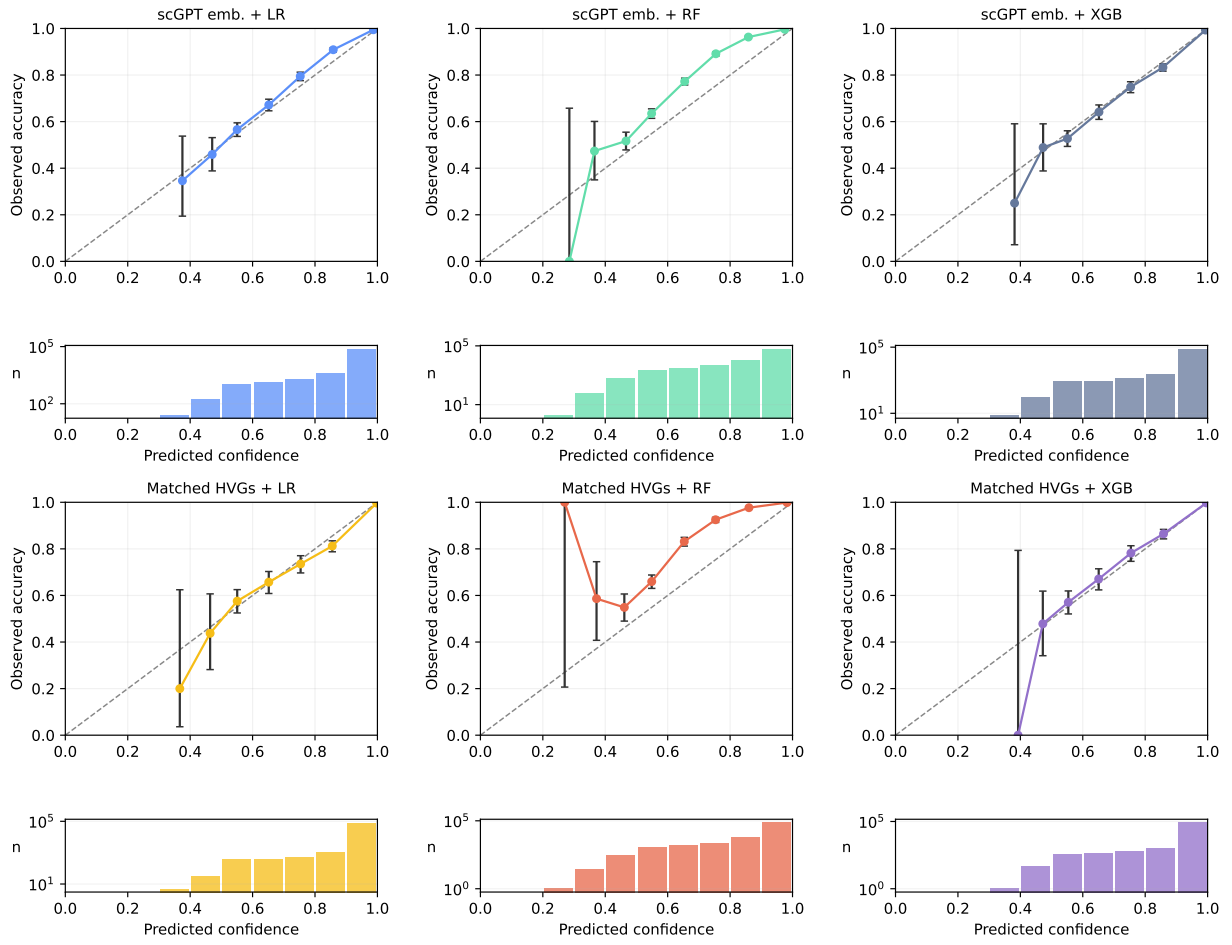
Supplementary Table S8: Computational environment and identity audit.

Component	Recorded value
Python	3.11.10
scGPT	0.2.5; source commit <code>cebd6fa</code>
PyTorch / CUDA	2.5.1+cu124 / 12.4
GPU	NVIDIA H100 80 GB HBM3
Scanpy / AnnData	1.11.5 / 0.12.3
scikit-learn / XGBoost	1.9.0 / 3.2.0
NumPy / SciPy / pandas	1.26.4 / 1.13.1 / 2.3.3
scGPT <code>args.json</code> SHA-256	<code>c18e075e018140cb8b2d9029387b9de26607a5ce6a8ccabd6ead70cd76b95d60</code>
scGPT <code>vocab.json</code> SHA-256	<code>acca93d114ca62c3f0f50debbd23e8c87f0714f4737764454f6b2b13f2e8580f</code>
scGPT <code>best_model.pt</code> SHA-256	<code>6cb5d451ab5c4b33eb673adbe4fddc61d2389df1b89b7651a9fe2e557572b922</code>

Supplementary reliability figures

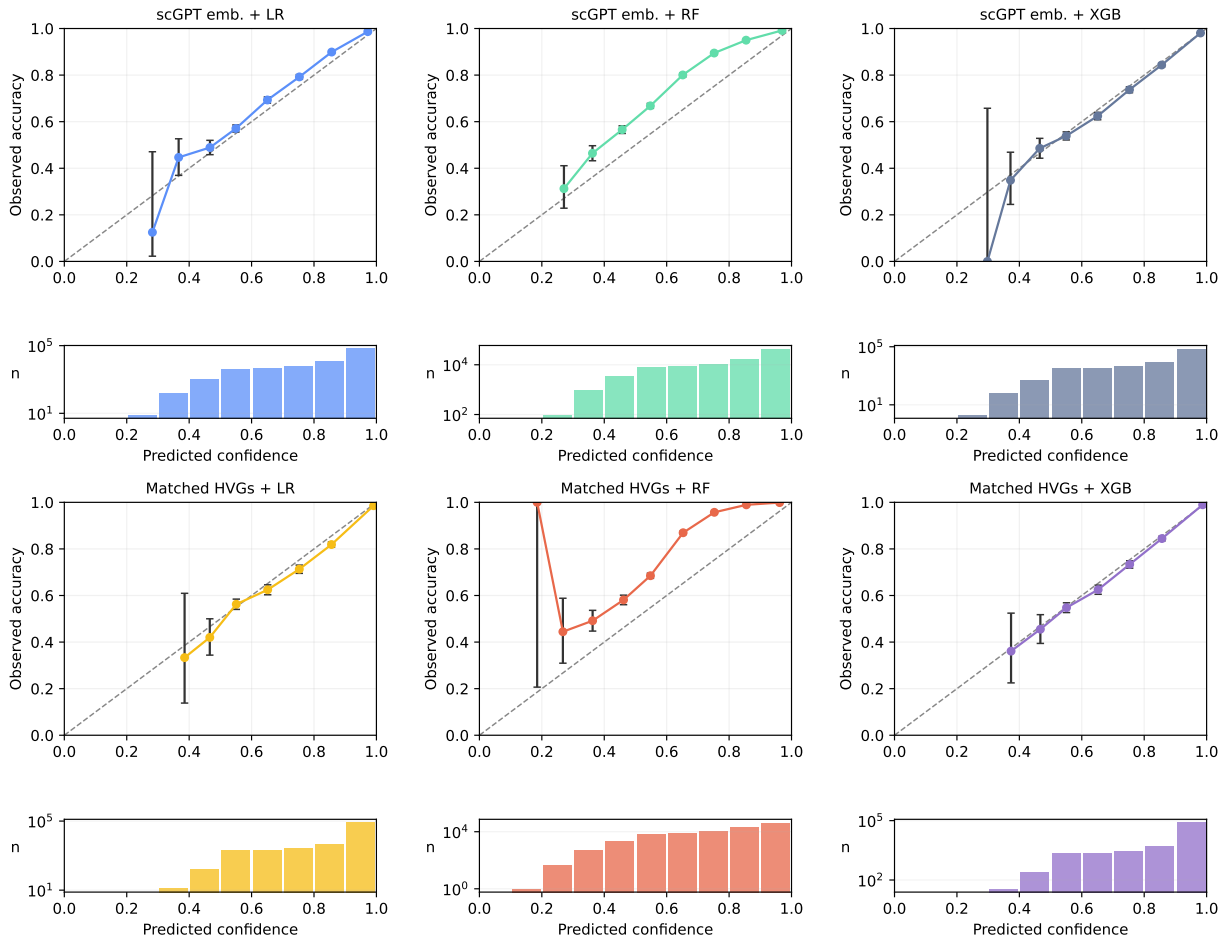
Every reliability figure uses ten fixed confidence bins. The upper panel for each pipeline plots observed accuracy against mean winning-class confidence; the lower panel reports the number of cells in each bin on a logarithmic scale. Within-atlas error bars are Wilson 95% intervals. Donor-held-out error bars are 95% donor-bootstrap percentile intervals. Empty bins remain visible as zero counts.

TNBC reliability with 95% Wilson intervals and bin counts



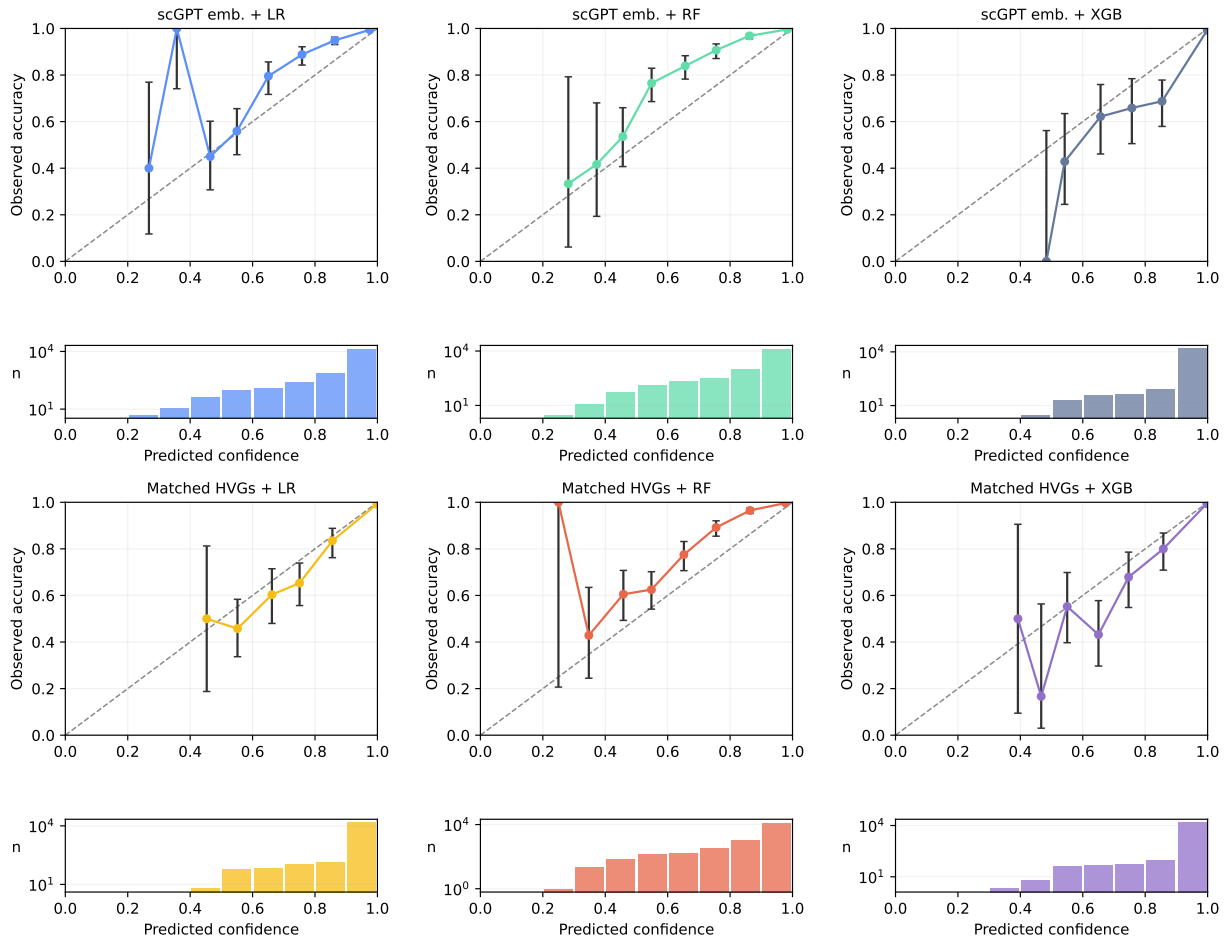
Supplementary Figure S1: Reliability of all six pipelines in the TNBC within-atlas test partition.

Indonesia PBMC reliability with 95% Wilson intervals and bin counts



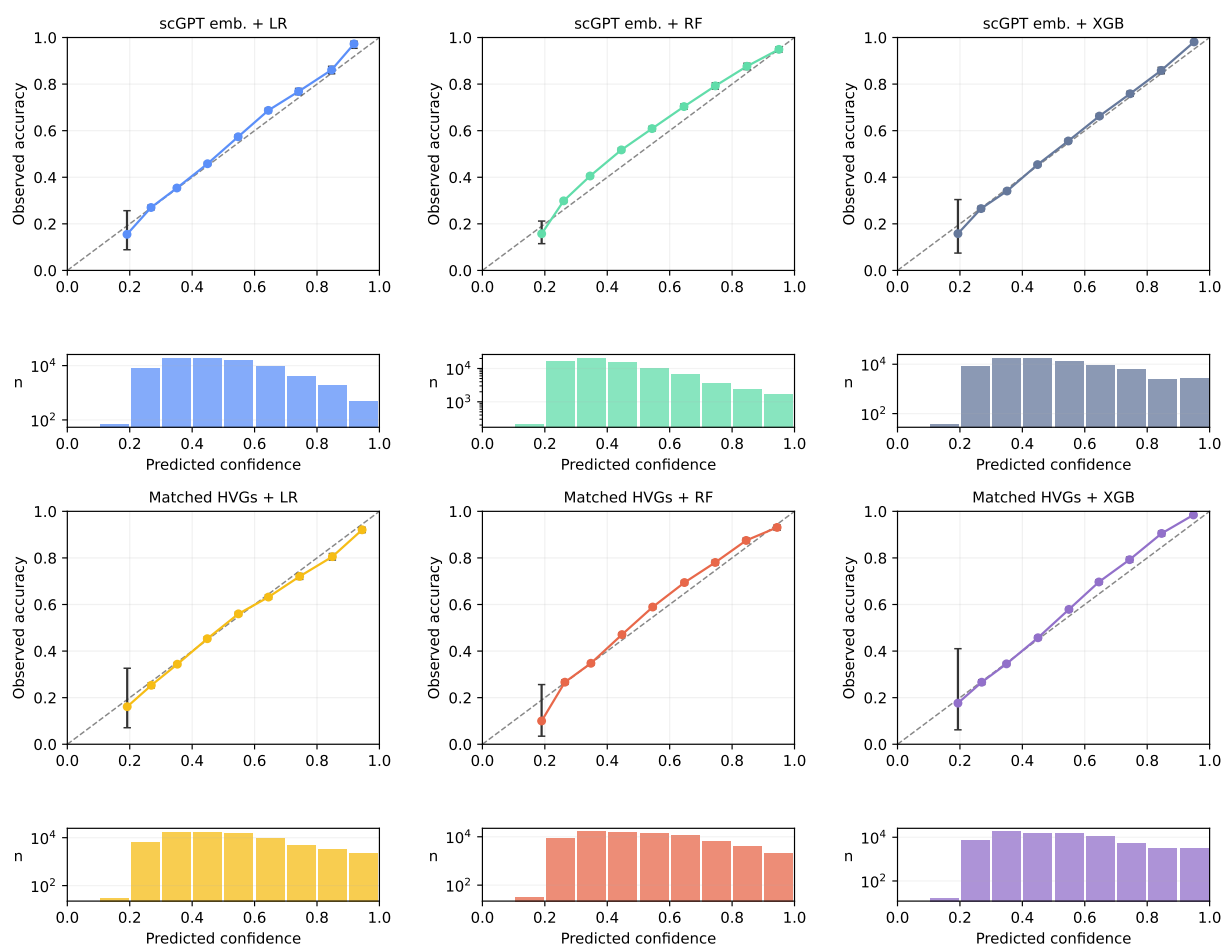
Supplementary Figure S2: Reliability of all six pipelines in the Indonesia PBMC within-atlas test partition.

Brain atlas reliability with 95% Wilson intervals and bin counts



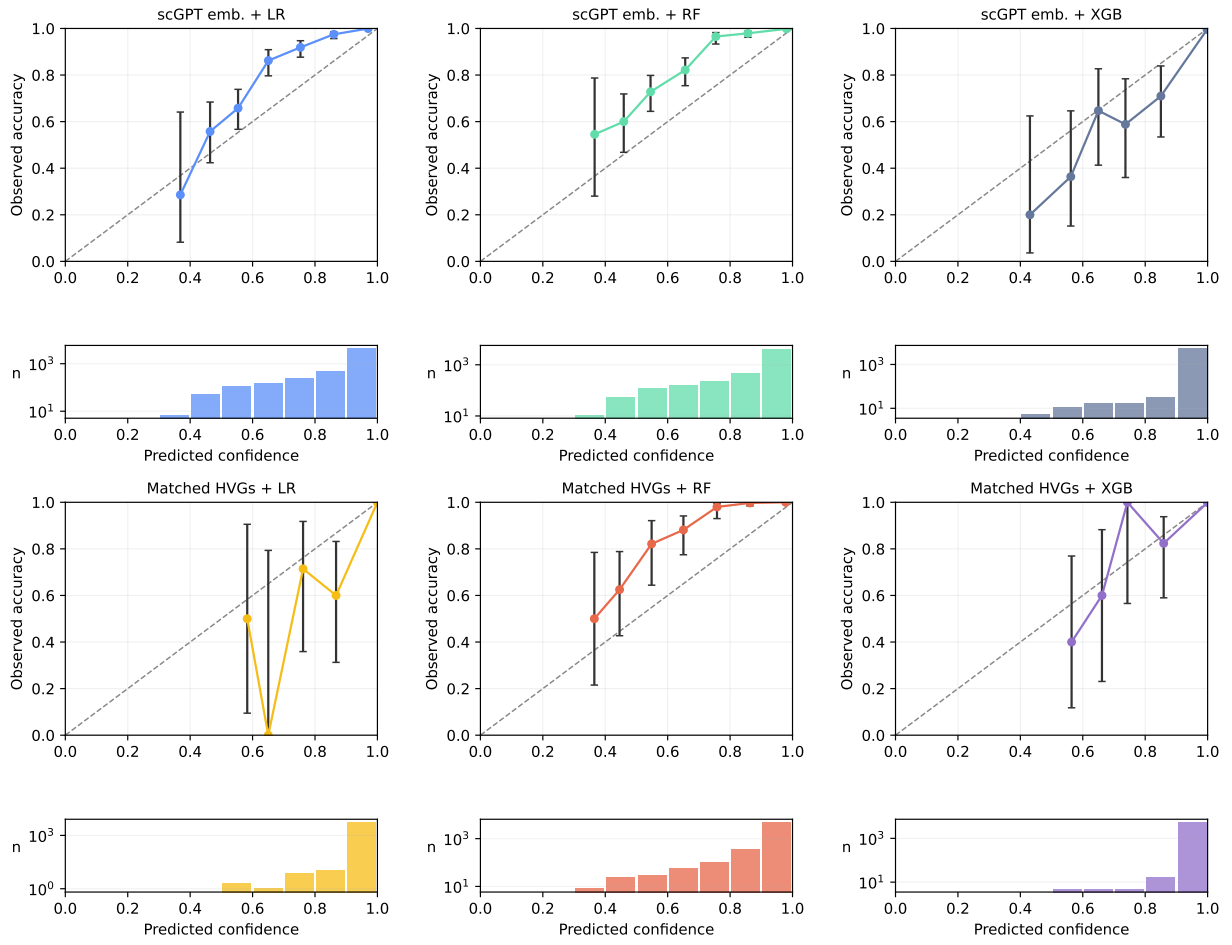
Supplementary Figure S3: Reliability of all six pipelines in the brain-atlas within-atlas test partition.

Multi-tissue TME reliability with 95% Wilson intervals and bin counts



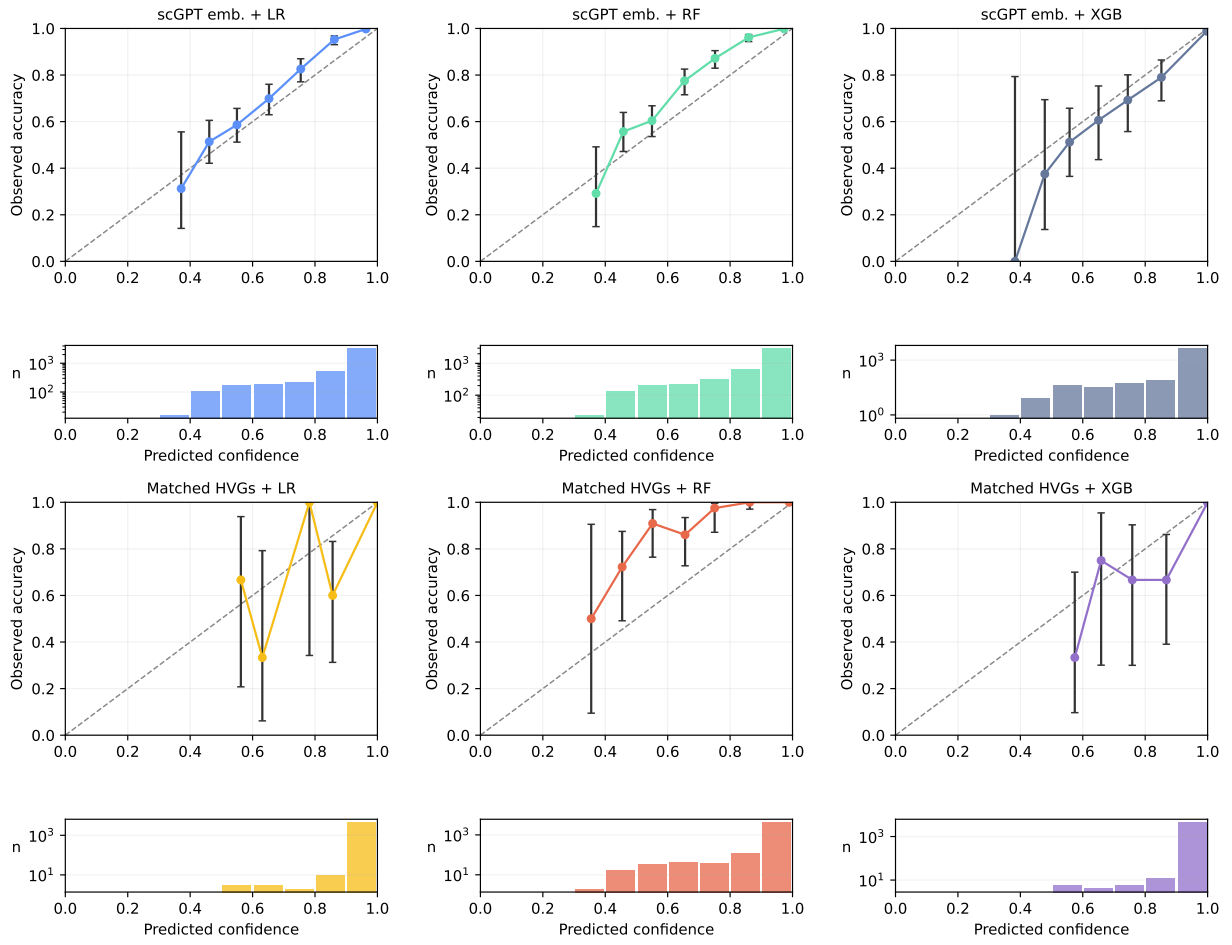
Supplementary Figure S4: Reliability of all six pipelines in the multi-tissue TME within-atlas test partition.

Human pancreas reliability with 95% Wilson intervals and bin counts



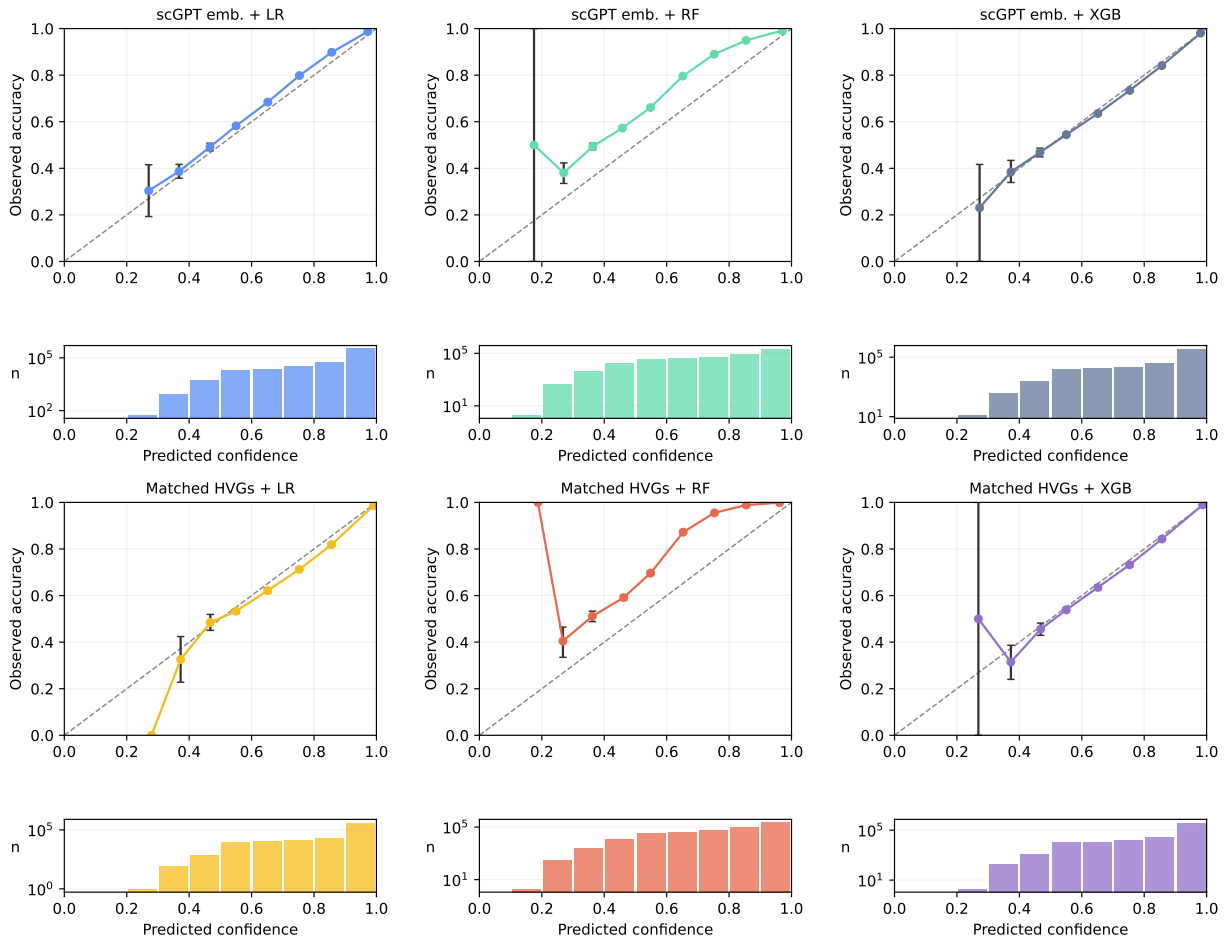
Supplementary Figure S5: Reliability of all six pipelines in the human-pancreas within-atlas test partition.

Pig pancreas reliability with 95% Wilson intervals and bin counts



Supplementary Figure S6: Reliability of all six pipelines in the pig-pancreas within-atlas test partition.

Donor-held-out reliability with 95% donor-bootstrap intervals and bin counts



Supplementary Figure S7: Reliability of all six pipelines from pooled out-of-fold predictions on 199 Indonesia PBMC donors. Accuracy intervals resample donors and carry all cells belonging to the selected donors together.